SEMESTER 5 · PROGRESS REVIEW · TMH OSTEONECROSIS AI PROJECT

Predicting osteoradionecrosis severity from dose and clinical data

A reproducible machine-learning pipeline that turns the MD Anderson ORN dataset (CSV) into graded risk predictions, explanations and a working research dashboard.

1,259
head and neck cancer patients

13.7%
developed ORN (grades 1–4)

0.74
test AUC for ORN yes / no

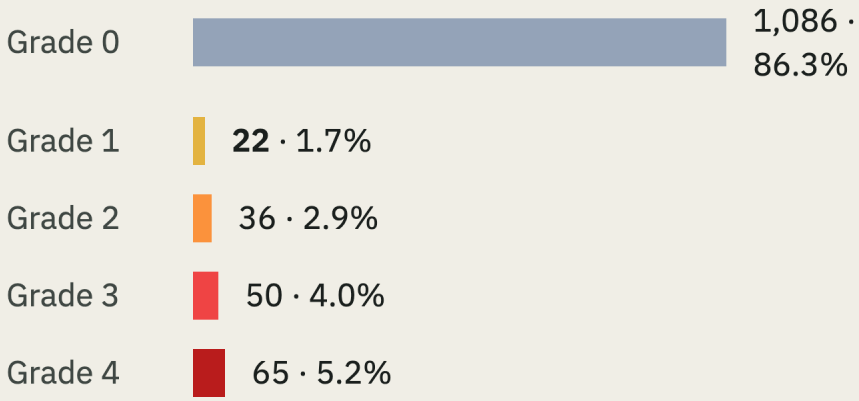
Muaaz, Vaishnavi, Kashish, Kamakshi · Guide: Vaishali Kulkarni· Tata Memorial Hospital, Mumbai · November 2026 · github.com/MuaazSM/orn-pipeline

From “will ORN occur?” to “how severe will it be?”

Published models predict ORN only as yes or no. We ask whether the same planning data (dose to the mandible and routine clinical factors) can also predict the **grade**, 0 to 4.

Study	Predicts	Best result
van Dijk 2021, NTCP	Yes / no	AUC 0.74
Reber 2023, ML vs DL	Yes / no	F1 0.31 (LR)
PREDMORN 2025	Yes / no	AUC 0.65–0.69
This work	Grade 0–4	Slide 6

ORN grade, MD Anderson cohort (n = 1,259)



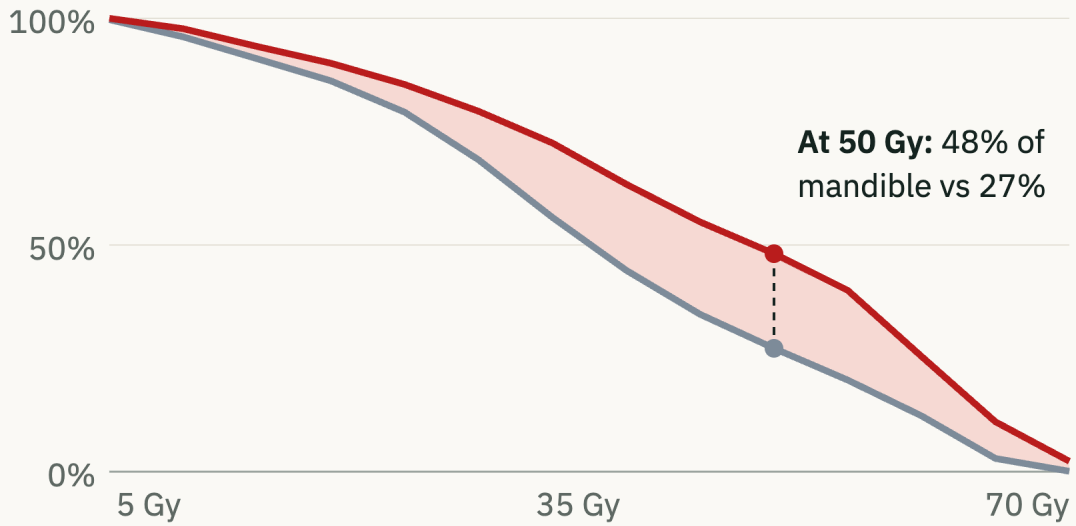
54 columns: demographics, smoking, T/N stage, chemo, dental extraction, mandible volume and full DVH (V5–V70 Gy, D2–D99%). Treated 2005–2015.

Data: van Dijk et al., Int J Radiat Oncol Biol Phys 2021 (figshare) · 2

02 · EXPLORATORY DATA ANALYSIS

Dose separates ORN from no ORN

Median mandible DVH — ORN (n = 173) — No ORN (n = 1,086)



ORN rate by clinical factor

Factor	Higher	Lower
Subsite	Oral 23.2%	Other 2.5%
T stage	T4 23.0%	T1 8.6%
Extraction	Yes 21.1%	No 11.0%
RT setting	Post-op 20.4%	Definitive 12.4%
Smoking	Current 15.0%	Never 12.7%

Cohort ORN rate 13.7%. Age, sex and mandible volume barely differ between groups.

Median D30: 58.5 Gy with ORN vs 48.1 Gy without

Missing data: only chemotherapy (20 patients), kept as “Unknown”

Redundancy: 224 of 780 DVH pairs at |ρ| > 0.85, so we built set B

Full cohort, n = 1,259 · ORN = grades 1–4 · DVH points are percent of mandible volume receiving at least that dose · 3

03 · PIPELINE

One command from raw CSV to a tested model

01
Load and validate
Schema, dtype and range checks; missing chemo kept as “Unknown”

02
Preprocess
ColumnTransformer: one-hot, ordinal T/N stage, scaler fit on train only

03
Split
Stratified 60 / 20 / 20 on grade, seed 42; indices saved

04
Feature sets
A: all 51 · B: 16 after |ρ| > 0.85 pruning · C: PCA on DVH

05
Cross-validate
5-fold CV over 47 configurations; SMOTE inside each fold

06
Select and test
Best macro-F1 refit on train + val; test set used once

07
Evaluate
1,000× bootstrap CIs, confusion matrix, ROC / PR, QWK

08
Explain
SHAP TreeExplainer, per-grade importance, partial dependence

SERVING joblib model + JSON artifacts → FastAPI: /predict, /explain, /results → Next.js research dashboard

python backend/run_pipeline.py --data data.csv · scikit-learn, imbalanced-learn, XGBoost, LightGBM, SHAP · 4

04 · MODELLING DESIGN

Four ways to frame the target, six models, three imbalance fixes

TARGET FORMULATION

Binary · ORN yes / no

3-class · none / mild (1–2) / severe (3–4)

Two-stage · ORN? then mild vs severe

5-class · grade 0 to 4

×

MODELS

Logistic regression (L2)

Random forest · XGBoost · LightGBM

Ordinal logistic (mord)

Stratified dummy baseline

×

IMBALANCE

None

Class weights

SMOTE, per fold

Why not deep learning

1,259 tabular rows. Reber et al. found LR beat ResNet on this data (F1 0.31 vs 0.07).

Selection metric

Macro-F1 weighs every grade equally; accuracy would reward predicting “no ORN”.

Sanity check

Replicating van Dijk’s 2-feature LR (D30 + extraction) gave CV AUC 0.78.

47 configurations scored by 5-fold CV on the training split · 5

05 · RESULTS ON THE HELD-OUT TEST SET (N = 252)

Whether ORN occurs is predictable; its grade is not, yet

BINARY · RANDOM FOREST

0.74

ROC AUC, 95% CI 0.66–0.81

Matches the published van Dijk NTCP benchmark (0.74). At a tuned threshold of 0.25: sensitivity 0.77, specificity 0.61.

Formulation	Macro-F1 [95% CI]	Chance	QWK
Binary	0.61	0.48	–
3-class	0.40 [0.33, 0.46]	0.36	0.22
Two-stage	0.42 [0.34, 0.49]	–	0.25
5-class	0.21 [0.18, 0.24]	0.20	0.17

5-class model, F1 per grade: it collapses to grades 0 and 4

Grade 0	Grade 1	Grade 2	Grade 3	Grade 4
0.89	**0.00**	**0.00**	**0.00**	**0.14**
n = 218	n = 4	n = 7	n = 10	n = 13

CIs: 1,000 bootstrap resamples · Chance = stratified dummy, CV macro-F1 · QWK = quadratic weighted kappa · 6

Mid-range mandible dose dominates the model

Top 8 features, mean |SHAP|



D30 is the same predictor van Dijk and PREDMORN identified. Correlated DVH points share credit, so read these as one dose band.

Takeaway

ORN risk is learnable from planning data; severity needs more cases or new signal.

Next semester

Add CT imaging features from the TMH cohort

Pool multi-centre data to grow grades 1–3

External validation and calibration

Code and data: github.com/MuaazSM/orn-pipeline